Final Project: Pick a Project

Computer Science - Week 10

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Please make sure to your UD **email** AND **name** is below. Unless your UD emails are included in this table, then you will not earn any points for this assignment when it is graded! **Every person should submit their own individual worksheet**.

Talk about each of the questions with a friend or our LLM (or both), but **make sure the words you enter below are yours** (with the obvious exceptions notwithstanding). Brainstorming is best done in full sentences, and the main goal here is brainstorming. It's worth reviewing the information in this series of pages in the Drafter documentation to gain a better understanding of the process.

Role	Name	Email
Developer (You)	Maximo Cordrey	maximo@udel.edu
Friend (optional)		

Part 1: The Goal & Prior Implementations

1. **Identify the Goal:** What specific goal are you trying to address? Think about your passions, hobbies, and academic interests. The latter are an excellent candidate to focus on. Some majors have suggestions listed below: these were provided by faculty in those departments. Other majors could consider a focus tied to a big final assignment in one of their recent courses in their major.
    - **Biomedical Engineering** majors may consider one of the following:
        - If you have started (or completed) the Cell/Tissue Design final project in BMEG100, design an organoid, which involves combining a cell type, biomaterial, and testing method. Have the website include a description and diagram of your design concept, a testing method, a needs statement, and an impact statement relevant to their organoid. Also include a link to a paper that supports your idea and a paragraph describing how that paper supports your design concept.

        - Otherwise, create a website related to your BMEG100 Lab Report 1, *Cytoskeletal and Nuclear Staining Lab Report Guidelines*, linked here (University of Delaware account required to access).
    - **Chemistry** and **Chemical Engineering** majors may consider creating an automated chemistry assistant that takes balanced reaction equations, starting reactant masses, and a harvested product amount to determine the limiting reactant, theoretical yield, and percent yield based on principles from Chemistry 2e (OpenStax) — Chapter 4: Chemical Reactions and Stoichiometry. To avoid manually compiling atomic mass tables or writing custom string tokenizers, use the ChemPy library to parse molecular formulas into exact molecular weights and verify reaction stoichiometry against compound data from the NIH PubChem Database. The program outputs an organized breakdown of limiting reagent status, product yields, and remaining masses of unreacted species. Also use the heat of the reaction to determine if it is exothermic or endothermic, and compute the heat flow per mole of the product.
    - Majors in the **Electric & Computer Engineering** department may consider a steganography project that hides messages in images, and then retrieves those messages from uploaded images.
    - **Mechanical Engineering** majors may consider a project that somehow uses both the NumPy and SciPy Python libraries.

My goal is to make a website that focuses on something construction related since my major is construction engineering.

2. **Prior Implementations:** Search the internet. What existing tools or websites currently attempt to address it?

There are a few websites that say they can build estimates in minutes. The ones I found you have to pay for so I was not able to test them out. One is called “Buildertrend” and another is called “STACKct.com.” This one uses AI to do the estimating and it says it is free.

3. **Your Implementation:** Briefly describe the legitimate purpose or functionality of your site. How would you want this web application to be used?

The purpose of this website will be to take the user inputted project and return them an estimate on how much of each material they will need to build this project, an estimated time to complete the project, and estimate the cost of materials using prices given by the user.

Part 2: The Input-Process-Output (IPO) Model

Think about the fundamental flow of your program.

1. **The Core Goal:** Every program has three main phases: taking input, processing data, and presenting output. In one sentence, what is the ultimate goal your program will accomplish for the user? Be specific and use CS1 concepts.

The ultimate goal is to take the users input of what the project is and the inputted cost of the materials process that and return them what materials they will need/how much of each and the total cost of the project.

2. Input: What specific information will the user need to provide to your program? (e.g., a city name, a numerical budget, a list of hobbies).

They will need to provide what type of project they are working on: a house, a shed, a garage, etc. After the materials are returned they will need to input the price of each material so the total cost of the project can be calculated.

3. Processing: At a high level, how will the computer manipulate or evaluate that data? (e.g., calculate the average, filter out options that are too expensive, compare the input against a database).

To decide what materials and how many it will likely compare the input against a database. Then using the sum of the cost of all the materials using the cost times the number of that material then adding up the total of each material.

4. Output: What will the user get back? (e.g., a recommendation, a calculated safety factor, a formatted schedule).

The user will get back the total cost of their project. Either as one number or the total cost for each material on its own.

Part 3: Data Abstraction

Identify the "nouns" of your problem.

1. **Representing Reality:** Abstraction is the process of representing the real world inside a computer by ignoring unnecessary details. What are the core "things"

(nouns) your program needs to track? (e.g., *If building a recipe app, the nouns are "Recipe" and "Ingredient"*).

Materials, Cost, Project, Price.

2. **Identifying Properties:** For the most important "thing" you listed above, what 3 to 4 specific details do you actually need to solve your problem, and what details will you abstract away?
    - *Details I need:* (e.g., *Ingredient name, weight, cost*)
    - *Details I will ignore:* (e.g., *Ingredient color, brand, expiration date*)

I need the Materials name, cost, and amount. I will ignore the Materials color, brand.

Part 4: Challenges, AI Strategy, & Goals

1. **Difficulties:** What parts of this project seem the most difficult or technically complex?

The hardest part will likely be assigning values to the inputted materials and finding a way to only let specific materials be entered and specifying that.

2. **Coding Strategy:** If you will be using an LLM to help code, how will you ensure you understand the generated code and avoid "hallucinations" or "shallow learning" (copying without understanding)? If you won't be using an LLM to help code, this project might take about 4 hours a week to complete, so list each of the days and times in which you work on the project.

When using an LLM to help me code at first I will not ask it directly to generate me code if I am stuck. Instead I will explain my issue and ask for tips on how to get around it or do whatever it is I am trying to do.

3. **Goals:** What is your ultimate goal for this project? (e.g., *Impress the graders with a creative UI, build something I can use in my other classes, learn Drafter deeply*).

My goal with this project is to better understand drafter and all the possibilities there are

when using it. I have not really understood it so far.

Part 5: Research & Brainstorming Conversation

Have a conversation with a Generative AI tool (like ChatGPT or Claude) OR a friend about your project idea. Do not ask for code. Instead, use the AI/friend as a sounding board to research your problem.

1. **The Initial Prompt/Question:** Ask the AI or your friend to help you brainstorm. (e.g., *"I want to build a simple web tool to help students track their study hours. What are 3 common problems students face when tracking their time, and what are 3 simple features a basic tool should have?"*). Paste your prompt below:

I want to build a simple web tool that helps people estimate the cost of their small projects quickly. What are some issues that people face when estimating? What are a few features the web tool should have to make it useful and worthwhile?

2. **The Response:** Briefly summarize the best ideas the AI or your friend suggested. What existing solutions or prior apps did they mention?

The AI said one of the issues is that sometimes people forget the price of small things like screws, nails, fasteners, etc. One feature it said I should have is a “material cost calculator” which is what I wanted to have. It also mentioned adding labor costs into it which would be a good idea.

3. **The Pushback:** You must reply to the AI/friend to narrow the scope when it's over-ambitious. Pick the most complex feature they suggested and explain why it is too complicated for an introductory programming project. Ask them for a simpler alternative. *Paste your response here:*

A feature it said to do was have an “automatic project cost breakdown” function. This may be too complicated because I would have to assign each specific material to a category. It also mentioned the cost of permits which would probably be different for every area. I asked it for a simple alternative and it said “A simpler alternative would be a Total Project Cost Calculator. Instead of automatically separating expenses into multiple categories, users simply enter three amounts: 1. Material costs, 2. Labor costs, 3. Additional expenses The website automatically adds them together to display one estimated project total.” This kind of defeats the purpose of the website as it requires the user to calculate the totals of each category.

4. **Reflection:** Based on this conversation, how has your project idea changed or become more focused?

One way my project became more focused because of this conversation is to look into more costs than just material costs. There are labor costs, permit costs, etc. It also recommended adding a section where the user can input their budget to see if the project is able to be done.

Part 6: Resource Planning

1. What resources will you use if you get stuck?

If I get stuck I will use an LLM, go to a friend, go to office hours, or ask in the class discord.

2. If your LLM provides code that causes a Drafter or Bakery error, or if you run into a problem during deployment, what is your step-by-step debugging plan to resolve it?

If this were to happen I would try to see if I can figure out what is causing the error on my own. If I am not able to figure it out I would attend office hours or ask a friend.

Part 7: LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

For my project I want to create a website that correlates with construction since my major is construction engineering. can you give me some possible concept suggestions

Since your major is construction engineering, I think you could build a website that solves a real construction-related problem while showing off your programming skills. The key is to choose something that is useful in the construction industry but still manageable for a CS1 project using Python and Drafter.

For your project, I'd focus on an application that takes user input, processes it through calculations or comparisons, and produces a useful result. That also fits the Input-Process-Output (IPO) model in your project planning worksheet.

7 construction engineering website ideas

1. BuildCalc — Construction Material Calculator

Best overall choice

A website that helps users estimate how much material they need for a small construction project.

Features

- Calculate concrete needed for a slab, footing, or column.
- Estimate the number of bricks, blocks, or flooring tiles.
- Estimate material costs using user-entered prices.
- Display a summary of quantities and estimated expenses.

Example: A user enters the length, width, and depth of a concrete slab, and the website calculates the required concrete volume.

Difficulty: Easy–Medium · Skills: Math, functions, input validation

I want to build a simple web tool that helps people estimate the cost of their small projects quickly. What are some issues that people face when estimating? What are a few features the web tool should have to make it useful and worthwhile?

For your web tool, I would recommend creating a Small Project Cost Estimator designed for homeowners, students, DIY builders, and small contractors. The goal would be to help someone quickly estimate how much a construction or renovation project will cost without having to manually research every material and labor expense.

Here are some common problems people face and features that could make your website useful.

1. Common problems people face when estimating project costs

1. Difficulty calculating material costs

- People often don't know how much material they need.
- Material prices vary depending on suppliers and locations.
- It's easy to forget smaller materials like screws, nails, adhesives, and fasteners.

2. Underestimating labor costs

- People may not know how many hours a project will take.
- Labor rates differ depending on the type of work and contractor.
- Unexpected complications can increase labor expenses.

3. Forgetting additional expenses

- Delivery fees, equipment rentals, permits, and disposal costs are often overlooked.
- People sometimes forget to account for unexpected expenses.

4. Difficulty comparing project options

- It can be difficult to determine whether one material or construction method is more affordable.
- People may struggle to compare doing the work themselves versus hiring a contractor.

5. Lack of construction estimating experience

- Beginners may not understand how to calculate quantities, measurements, or total project expenses.
- Many estimating programs are too complicated for someone completing a small project.

2. Features your web tool should have

Here are some features I would consider incorporating into your website.

My Designer Plus — Design. Estimate. Quote.

1. Material Cost Calculator

Allow users to select materials, enter quantities, and automatically calculate total material costs.

Example: 20 bags of concrete at $7 per bag = $140.

Construction Worker Measuring Wood Beam with Tape Measure

2. Labor Cost Estimator

Let users enter estimated working hours and hourly labor rates to calculate labor expenses.

Rakentamisen budjetti ja kustannusarvio | Miten välttää yllätykset?

3. Automatic Project Cost Breakdown

Automatically organize estimated expenses into categories:

- Materials
- Labor
- Equipment
- Permits and miscellaneous expenses
- Contingency allowance

Horux Construction | Transform Your Home Today – Get a Quote

4. Project Templates

Allow users to select common small projects instead of starting from scratch.

Examples:

- Building a deck
- Installing a fence
- Remodeling a bathroom
- Installing drywall
- Pouring a concrete patio

ChatGPT for Construction Estimates | TaskTag

5. Budget Comparison

Allow users to enter their available budget and compare it against their estimated project cost.

Estimate Template with Labor & Material Breakout (Excel) – Designs By Blue Star

6. Downloadable Cost Estimate

Allow users to download or print a report containing their project details, material quantities, labor expenses, and estimated total.

Part 8: Submit

In Google docs, with this document open, please click on *File* > *Download* > *Microsoft Word (.docx)*, and make sure you put that download somewhere you can find it again. Then, upload it to the Canvas assignment about selecting or brainstorming your final project.

When you're done with this assignment, review this page of the Drafter documentation to better prepare for the next step.